TAMBAN THESIS

Defense Survival Pack & System Source of Truth

Evidence-backed reference for architecture, claims, grading policy, final evaluation, limitations, and defense answers

Prepared for	Tamban dried-fish computer vision thesis project
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Status	Working defense source of truth - update after expert validation and final test

Use this document to keep every team member aligned on what the system actually does, what is proven, what is still a hypothesis, and what must never be overclaimed during defense.

How to Read This Defense Pack

This document separates implementation facts from internal-development evidence, expert-dependent rules, and final-test claims. The distinction is intentional: a strong defense is not about pretending every component is perfect; it is about knowing exactly what is implemented and what the evidence supports.

Status	Meaning	Defense use
IMPLEMENTED	Present in the current system/code or decision policy.	Safe to describe as system behavior, but do not imply validated accuracy unless separately tested.
INTERNAL EVIDENCE	Supported by development/engineering data.	Safe only with the qualifier “on the internal-development split/experiment.”
EXPERT PENDING	Needs source-of-truth annotator/domain expert confirmation.	Present as a design hypothesis or proposed policy, not an industry standard.
FINAL TEST PENDING	Needs specimen-independent end-to-end validation.	Do not call it final accuracy or production reliability yet.
PROPOSED	Planned or optional enhancement, not active baseline.	Say “we are evaluating / propose,” never “the system already does.”

Defense Ground Rules

• Never call detector mAP “final grading accuracy.”

• Never treat a missing Head or Tail detection as proof that the part is physically missing.

• Never say the 50/30/20 weights are scientifically established until expert elicitation and ablation are completed.

• Never say EfficientNet is inherently unsuitable; the tested EfficientNetV2-M + RetinaNet configuration failed in this experiment.

• Never call current development metrics specimen-independent final results.

• Never claim handcrafted color/texture/geometry features improve the final verdict unless an ablation demonstrates improvement.

• When evidence is insufficient or ownership is ambiguous, abstention (Ungraded/Unassigned) is an intentional safety behavior, not a fake grade.

1. System Source of Truth

The exact story every team member should tell about the system during defense.

1.1 One-Sentence System Description

The thesis system is a multi-stage computer-vision pipeline that first localizes individual dried tamban, then detects grade-aware Head/Body/Tail evidence, associates that evidence with the correct parent fish, and applies a conservative rule-based grading policy that can return A, B, C, Rejected, or Ungraded.

1.2 Component Responsibilities

Component	Primary job	What it does NOT prove	Current status	Defense sentence
Model 1	Detect/localize candidate whole fish; provide parent region/identity and support counting/tracking.	It does not decide A/B/C/Rejected quality.	IMPLEMENTED	"Model 1 answers where the fish is and provides the parent fish region."
Model 2	Detect 12 grade-part classes: A/B/C/Rejected × Head/Body/Tail.	A part detection by itself is not yet a whole-fish verdict.	IMPLEMENTED + INTERNAL EVIDENCE	"Model 2 supplies anatomical quality evidence, not the final fish grade by itself."
Part association	Assign each Model 2 part to the correct Model 1 parent; allow ambiguous/unassigned evidence.	It is not automatically correct just because boxes overlap spatially.	IMPLEMENTED/UNDER VALIDATION	"Ownership is a separate problem and ambiguous parts should not be forced to a fish."
Grading engine	Combine reliable region evidence using original Body/Head/Tail weights; apply evidence gates and abstention.	Weighted support is not a calibrated probability.	IMPLEMENTED; POLICY VALIDATION PENDING	"The final verdict is rule-based evidence fusion on top of detector outputs."
Handcrafted features	Measure color, brightness, shine, texture and geometry for analysis/possible fusion.	Their presence in CSVs does not mean they currently improve the final grade.	RESEARCH / ABLATION PENDING	"We only claim a feature helps if ablation shows measurable benefit."
Fish verifier	Potential second-stage Fish/Not-Fish check for Model 1 candidates.	It is not part of the baseline until trained and evaluated.	PROPOSED	"The verifier is a planned false-positive mitigation experiment, not a completed capability."
Segmentation	Potential exact fish masking for background-resistant features/shape.	Segmentation does not automatically solve false positives and is not required for the current baseline.	OPTIONAL / PROPOSED	"We would add segmentation only if measured errors justify the added complexity."

1.3 Current Model 2 Internal-Development Evidence

These values are useful for architecture comparison only. They must not be presented as final specimen-independent fish-level grading accuracy.

Model	mAP50-95	mAP50	Precision	Recall	F1	Latency ms/img	Interpretation
YOLO26m	0.4606	0.7876	0.8018	0.8625	0.8310	54.6	Strongest internal-development AP/latency profile; current lead candidate, not final winner.
Faster R-CNN R50-FPN V2	0.4405	0.7665	0.8122	0.8553	0.8331	123.7	Comparable detector quality; slightly higher F1 at its own tuned threshold, but slower.
EfficientNet V2-M + RetinaNet	0.0000	0.0000	0.0000	0.0000	0.0000	140.2	The tested configuration failed to produce usable evaluated detections; diagnose configuration rather than claiming EfficientNet itself is invalid.

1.4 Model 2 Class Map

Grade_A_Body	Grade_A_Head	Grade_A_Tail
Grade_B_Body	Grade_B_Head	Grade_B_Tail
Grade_C_Body	Grade_C_Head	Grade_C_Tail
Rejected_Body	Rejected_Head	Rejected_Tail

Known internal-development weakness: tail classes, especially Grade_B_Tail, are materially weaker than the strongest body/head classes. Treat missing tail evidence conservatively.

1.5 Grading Semantics

Current design weights	Body = 50%, Head = 30%, Tail = 20%.
Missing evidence	Do not renormalize the remaining regions back to 100%. Missing or unreliable evidence contributes zero support and lowers evidence coverage.
Weighted support	For grade g: Score(g) = 0.50×Body_conf(g) + 0.30×Head_conf(g) + 0.20×Tail_conf(g), using only reliably associated evidence and retaining the original weights.
Evidence coverage	Coverage = 0.50×I(Body reliable) + 0.30×I(Head reliable) + 0.20×I(Tail reliable).
Rejected	Positive evidence of unacceptable/rejected-quality appearance or a validated rejection rule. Not synonymous with missing detection.
Ungraded	Insufficient, conflicting, ambiguous, occluded, out-of-frame, or otherwise unreliable evidence. It is an abstention state, not a low-quality class.
Body-only policy	NOT YET SCIENTIFICALLY FROZEN. Expert validation and a gating ablation should decide whether Body-only evidence can ever issue an automatic grade.

1.6 Current Integration Story

Keep the integration story precise:

1. Original image/frame
2. Model 1 finds candidate parent fish
3. Model 2 detects grade-aware Head/Body/Tail evidence
4. Association links each part to at most one parent fish; ambiguous evidence may remain UNASSIGNED
5. Reliable evidence and optional engineered features are summarized per fish
6. The grading policy computes non-renormalized weighted support and applies uncertainty/rejection gates
7. Output is A / B / C / Rejected / Ungraded, with Fish ID and evidence trace

1.7 What Is Not Yet Proven

☐ Final specimen-independent fish-level grading accuracy.

☐ That 50/30/20 is the optimal or expert-approved weighting.

☐ That the system generalizes reliably to arbitrary backgrounds.

☐ That overlap association is accurate enough under heavy occlusion without a dedicated benchmark.

☐ That handcrafted appearance/geometry features improve final grading.

☐ That a Fish/Not-Fish verifier reduces false positives without rejecting too many real fish.

☐ That a missing anatomical detection corresponds to a physically missing part.

2. Claim → Evidence Matrix

Use this table when a panelist asks: “What is your proof?”

A claim should be no stronger than its evidence. “Internal” means the evidence is useful for development but should not be generalized to final unseen specimens.

Claim	Current status	Evidence available now	What is still missing	Safe defense wording	Priority
Model 1 can localize individual fish.	**INTERNAL EVIDENCE**	Model 1 training/stress-test outputs and live observations.	Independent target-environment detector test with hard negatives/background subgroups.	“Model 1 is our parent-fish detector; current performance evidence is developmental, and final detector generalization is tested separately.”	**P0**
Model 2 learned the 12 grade-part classes.	**INTERNAL EVIDENCE**	YOLO26m/Faster R-CNN internal-development AP, precision, recall, per-class results.	Independent specimen-level test and end-to-end fish verdicts.	“Both YOLO26m and Faster R-CNN learned the part-detection task on the internal development split.”	**P0**
YOLO26m is better than Faster R-CNN.	**PARTIAL / INTERNAL**	Higher internal mAP50-95/mAP50 and ~2.3× lower measured latency; F1 is nearly tied.	Independent same-test-set comparison.	“YOLO26m currently has the strongest internal AP/latency profile; final selection remains subject to independent testing.”	**P1**
EfficientNet is unsuitable.	**DO NOT CLAIM**	Only the tested EfficientNetV2-M + RetinaNet configuration produced zero usable metrics.	Debug/re-run if architecture-level conclusion is desired.	“That configuration failed; we do not generalize the failure to EfficientNet as a family.”	**P1**
Body/Head/Tail should be weighted 50/30/20.	**EXPERT + EXPERIMENT PENDING**	Current design hypothesis and implemented policy.	Independent expert elicitation + weight ablation on	“50/30/20 began as a design hypothesis and will be retained	**P0**

Claim	Current status	Evidence available now	What is still missing	Safe defense wording	Priority
			validation data.	only if expert and validation evidence support it.”	
Missing part evidence should not be renormalized.	**IMPLEMENTED / CONCEPTUALLY SUPPORTED**	Current conservative grading policy; avoids converting partial evidence into full evidence.	Expert confirmation of missing-evidence handling; coverage threshold validation.	“Unavailable evidence remains unavailable; we do not manufacture confidence by redistributing its weight.”	**P0**
Body alone is enough to grade.	**UNRESOLVED**	Current software can reach the body weight ceiling; policy still requires expert/validation decision.	Annotator scenario questions + gating ablation.	“We have not frozen a Body-only grading rule; this is explicitly being validated.”	**P0**
Ungraded is different from Rejected.	**IMPLEMENTED POLICY / EXPERT PENDING**	Current decision semantics and safety design.	Domain-expert confirmation of rejection conditions.	“Rejected means positive bad-quality evidence; Ungraded means insufficient trustworthy evidence.”	**P0**
Part ownership is reliable when fish overlap.	**NEEDS EXPERIMENT**	Association implementation/prototype supports ambiguity and unassigned outcomes.	Parent-ID ground truth and overlap benchmark.	“The system can abstain on ambiguous ownership; association accuracy itself must be measured on reviewed parent IDs.”	**P0**
The system works on different backgrounds.	**NOT YET PROVEN**	Observed strong conveyor performance and known false positives outside it.	Background-generalization benchmark; hard-negative/verifier experiment.	“Background generalization is a known limitation being measured rather than assumed.”	**P0**
Handcrafted features improve grading.	**NEEDS EXPERIMENT**	Color/shine/texture/geometry features are extractable; legacy feature artifacts exist.	Model 2-only vs +geometry vs +appearance vs +both ablation.	“These are candidate explanatory features until ablation shows incremental value.”	**P1**
B and C are reliably distinguishable.	**NEEDS EXPERIMENT**	Internal Model 2 results show learning but historical B/C capture confounds	Mixed B+C scenes under shared lighting/background, new physical fish.	“B/C discrimination is one of the critical validation targets, particularly under controlled shared	**P0**

Claim	Current status	Evidence available now	What is still missing	Safe defense wording	Priority
		exist.		conditions.”	
The dataset has no leakage.	**NOT YET PROVEN AT SPECIMEN LEVEL**	Some source-group/duplicate audits exist.	Physical specimen IDs/session-safe split and final manifest.	“Image-level duplicate checks are not equivalent to specimen independence; final splits are specimen-aware.”	**P0**
The system is real-time.	**NEEDS TARGET-HARDWARE TEST**	Model-level latency exists for Model 2.	End-to-end latency including Model 1, association, UI, verifier if used, and target hardware.	“We report measured component latency separately from end-to-end system throughput.”	**P1**
The verifier reduces false positives.	**PROPOSED**	Architecture concept only.	Train/evaluate Fish vs Not-Fish verifier and measure retained fish recall vs FP reduction.	“The verifier is a proposed mitigation until an ablation demonstrates net benefit.”	**P1**
The final system grades fish accurately.	**FINAL TEST PENDING**	Component-level internal evidence.	Locked specimen-independent end-to-end final test.	“Final fish-level accuracy will be reported only after the complete locked pipeline is tested on unseen specimens.”	**P0**

2.1 Evidence Binder Rule

For every claim used in the manuscript or defense slide, archive the following together:

☐ exact checkpoint filename + hash

☐ dataset/split manifest

☐ config/thresholds

☐ evaluation script version

☐ result CSV/graph

☐ date of run

☐ claim-safe wording

☐ whether the evidence is internal development or final test

3. 50/30/20 Expert Validation + Ablation Plan

Turn the current weighting from “we made it” into an expert-informed, empirically tested grading policy.

3.1 What 50/30/20 Means Today

Body = 50%, Head = 30%, Tail = 20% is currently a design hypothesis implemented in the grading policy. It must not be described as an industry standard, biological law, or statistically optimal rule until the following two validation layers are completed.

Layer	Purpose	Outcome
Expert elicitation	Establish whether experienced graders independently assign similar relative importance and missing-part rules.	Domain justification
Validation ablation	Compare weighting/gating alternatives on the same validation set before final testing.	Empirical justification

3.2 Expert Elicitation Protocol - Do Not Anchor the Expert

Do NOT begin by saying “we use 50/30/20; is that okay?” Ask independently, then reveal the current weights only after the expert has committed to their own reasoning.

☐ Q1. Rank Body, Head, and Tail from most important to least important for quality grading.

☐ Q2. Distribute exactly 100 importance points among Body, Head, and Tail.

☐ Q3. Explain what visual information each region contributes to A/B/C/Rejected decisions.

☐ Q4. If only the Body is clearly visible, would you issue a grade? Under what circumstances?

☐ Q5. If Body + Head are visible but Tail is hidden, would you grade? How confident would you be?

☐ Q6. If a Head/Tail is not visible, how do you distinguish hidden/out-of-frame/not-detected from physically missing?

☐ Q7. Would inability to see a part ever make the fish Rejected by itself?

☐ Q8. If Body and Head disagree on grade, which region dominates and why?

☐ Q9. Which visual conditions are automatic rejection defects regardless of otherwise good appearance?

☐ Q10. What is the hardest grade boundary (especially B vs C), and which cues resolve it?

3.3 Expert Weight Worksheet

Region	Expert 1	Expert 2	Expert 3 / Notes
Body	____ /100	____ /100	_____
Head	____ /100	____ /100	_____
Tail	____ /100	____ /100	_____

Total	100	100	100 / N/A

After independent answers are recorded, show the current 50/30/20 rule and ask whether the expert would retain, change, or condition it on defect type.

3.4 Two Separate Ablations - Do Not Confound Weighting and Missingness

A common methodological mistake would be to compare weights and missing-part gates in one mixed experiment. Use two ablations:

A. Weight ablation on cases with reliable Head + Body + Tail

This isolates the effect of the weights themselves because every region is present.

Variant	Body	Head	Tail	Purpose
W1 Current	0.50	0.30	0.20	Current hypothesis
W2 Equal	0.333	0.333	0.333	Neutral baseline
W3 Body-heavy	0.60	0.25	0.15	Tests stronger Body prior
W4 Expert	expert	expert	expert	Domain-informed weights
W5 Learned	validation constrained	validation constrained	validation constrained	Optional non-negative weights summing to 1; fit only on validation/training, never test

B. Evidence-gating ablation on the full validation set

This tests whether the system should require Body only, at least two regions, or all three regions before automatically grading. Keep the selected weight scheme fixed during this experiment.

Gate	Automatic grade allowed when	Why test it	Main risk
G1 Body-only permitted	Reliable Body passes score/margin gates	Maximizes coverage	May overgrade partial evidence
G2 ≥2 reliable regions	Body + at least Head or Tail	Balances coverage and anatomical support	Could still miss critical rejection evidence
G3 Body+Head required	Reliable Body and Head	Matches stronger 80% designed coverage	Tail-specific defects may be missed
G4 All H+B+T	All three reliable	Most conservative evidence completeness	High Ungraded rate because Tail is weakest

3.5 Metrics and Decision Rule

Primary metric	Macro F1 across A/B/C/Rejected on the validation set, because classes are imbalanced and all grades matter.
Coverage	Automatically graded fish / all detected eligible fish.
Selective error	Wrong automatic grades / automatically graded fish. Lower is better.
Ungraded rate	Ungraded / eligible fish. Report alongside accuracy; do not hide abstentions.
B vs C confusion	Track B predicted C and C predicted B separately because this is a known hard boundary.
Rejected false acceptance	True Rejected predicted A/B/C divided by true Rejected. This is an important safety/quality guardrail.
Decision rule	Choose weights/gates using validation only. Freeze them before opening the final test results. If differences are negligible, prefer the simpler expert-supported policy.

3.6 Ablation Results Sheet - Fill After Run

Variant	Macro F1	Balanced Acc.	B recall	C recall	Rejected recall	Coverage	Ungraded	Selective error
W1 50/30/20	___	___	___	___	___	___	___	___
W2 equal	___	___	___	___	___	___	___	___
W3 body-heavy	___	___	___	___	___	___	___	___
W4 expert	___	___	___	___	___	___	___	___
W5 learned	___	___	___	___	___	___	___	___
G1 body-only	___	___	___	___	___	___	___	___
G2 ≥2 regions	___	___	___	___	___	___	___	___
G3 Body+Head	___	___	___	___	___	___	___	___
G4 all H+B+T	___	___	___	___	___	___	___	___

3.7 Defense Answer After Validation

Target answer: "The original 50/30/20 rule was a design hypothesis. We first elicited the source-of-truth annotator's independent ranking and 100-point allocation, then evaluated the expert-informed weighting against equal and alternative weighting schemes on validation data. We froze the selected policy before the final test."

4. Final End-to-End Fish-Level Evaluation Sheet

This is the result that answers: "How often does the complete system grade an individual fish correctly?"

4.1 Freeze the System Before Final Test

Item	Frozen value	Verified by/date
Model 1 checkpoint + SHA-256	_____	_____
Model 2 checkpoint + SHA-256	_____	_____
Model 2 class map	_____	_____
Detector confidence/NMS	_____	_____
Association algorithm + thresholds	_____	_____
Grading weights	_____	_____
Evidence coverage gate	_____	_____
Winner support/margin gate	_____	_____
Rejected policy	_____	_____
Runtime commit/tag	_____	_____
Final test manifest	_____	_____

4.2 Final Test Dataset Requirements

☐ Every physical fish has a permanent specimen_id.

☐ No physical specimen appears in training, validation, and final test simultaneously.

☐ Capture sessions are documented; near-duplicate/burst frames are kept within one split.

☐ Ground truth contains final grade A/B/C/Rejected for every eligible test fish.

☐ For overlap scenes, ground truth includes parent Fish ID and Head/Body/Tail ownership.

☐ Visibility state is recorded where possible: PRESENT, OCCLUDED, OUT_OF_FRAME, NOT_DETECTED/UNKNOWN, POSSIBLY_MISSING only if supported.

☐ Include mixed B+C scenes under the same lighting/background to reduce session shortcut explanations.

☐ Include isolated, overlap, occlusion, edge-of-frame, target conveyor, and representative alternate-background cases.

☐ Do not tune thresholds after inspecting final-test outcomes.

4.3 Final Confusion Matrix Template

Ground-truth rows contain known quality classes. Ungraded appears as a prediction column and is reported as abstention, not counted as a correct class prediction.

Actual \ Pred.	A	B	C	Rejected	Ungraded
A	____	____	____	____	____
B	____	____	____	____	____
C	____	____	____	____	____
Rejected	____	____	____	____	____

4.4 Core Metrics

Metric	Formula / definition	Why it matters	Final value
Macro F1	Mean class-wise F1 across A/B/C/Rejected.	Does not let common classes dominate the result.	____
Balanced accuracy	Mean recall across A/B/C/Rejected.	Useful under class imbalance.	____
Automatic grading coverage	# A/B/C/Rejected outputs ÷ eligible fish.	Shows how often the system commits to a grade.	____
Ungraded rate	# Ungraded ÷ eligible fish.	Shows abstention burden.	____
Selective accuracy	Correct automatic grades ÷ all automatic grades.	How trustworthy grades are when the system chooses to grade.	____
Selective error	Wrong automatic grades ÷ all automatic grades.	Lower is better; complements coverage.	____
Rejected false acceptance	True Rejected predicted A/B/C ÷ true Rejected.	Important product-quality risk.	____
B vs C confusion	B→C and C→B counts/rates.	Measures the hardest	____

		boundary directly.	
Association accuracy	Correctly assigned parts ÷ reviewed assignable parts.	Separates ownership errors from detector errors.	____
Association coverage	Confidently assigned reviewed parts ÷ visible reviewed parts.	Shows how often association can safely act.	____
Model 1 missed-fish rate	Ground-truth fish missed ÷ ground-truth fish.	Parent detector bottleneck.	____
False fish per image	Non-fish Model 1 detections ÷ test images.	Quantifies false-positive problem.	____
Count MAE	Mean absolute error between true and predicted fish count.	Direct counting quality.	____
End-to-end latency	Camera/input → final fish records, on target hardware.	System practicality; not the same as Model 2 latency.	____ ms

4.5 Per-Class Result Sheet

Class	Precision	Recall	F1	Support	Ungraded count	Most common error
A	____	____	____	____	____	_____
B	____	____	____	____	____	_____
C	____	____	____	____	____	_____
Rejected	____	____	____	____	____	_____

4.6 Subgroup Stress Tests

Subgroup	N fish	Macro F1	Coverage	Association error	Notes
Isolated fish	____	____	____	____	_____
Mild overlap	____	____	____	____	_____
Heavy overlap	____	____	____	____	_____
Occluded parts	____	____	____	____	_____
Edge-of-frame	____	____	____	____	_____
Mixed B+C	____	____	____	____	_____

Target conveyor	____	____	____	____	_____
Alternate background	____	____	____	____	_____
Low/different lighting	____	____	____	____	_____

4.7 Error Attribution - Where Did the Final Mistake Come From?

Failure layer	Count	% of final errors	Typical example	Action
Model 1 missed fish	____	____	_____	_____
Model 1 false fish	____	____	_____	_____
Model 2 missed part	____	____	_____	_____
Model 2 wrong grade-part class	____	____	_____	_____
Association/ ownership error	____	____	_____	_____
Evidence gate/threshold error	____	____	_____	_____
Ground-truth/ annotation issue	____	____	_____	_____
Lighting/background shift	____	____	_____	_____
Unknown/other	____	____	_____	_____

4.8 Final-Test Sign-Off

☐ All weights/gates were frozen before final-test evaluation.

☐ No final-test result was used to retune thresholds.

☐ Specimen independence was verified from the final manifest.

☐ All excluded cases and exclusion reasons are documented.

☐ The reported metrics distinguish detector, association, grading and abstention performance.

☐ Final conclusions use fish-level evidence, not only detector mAP.

5. Limitations, Failure Handling & Hard-Question Cheat Sheet

Know the weakness, know the safe behavior, and know the evidence needed to defend it.

5.1 Failure-Handling Matrix

Situation	Main risk	Safe system behavior	What to measure/show	Defense position
Different background / clutter	Model 1 false positives or unstable boxes.	Restrict target ROI where appropriate; use hard-negative/background-diversity training; optional verifier if validated.	FP/image and fish recall by background.	Known domain-shift limitation; do not claim arbitrary-background robustness.
Two fish overlap	Head/Body/Tail may attach to wrong parent.	Allow AMBIGUOUS/UNASSIGNED; abstain when ownership is unreliable.	Association precision/coverage by overlap level.	Ownership is evaluated separately from detection.
Tail not detected	Could be occluded, out-of-frame, small, or missed.	Mark unavailable/unknown; do not infer physical absence automatically.	Tail recall + visibility labels.	Missing detection ≠ missing anatomy.
Only Body visible	Risk of overconfident partial grade.	Use expert/validation-chosen evidence gate; otherwise Ungraded.	Gating ablation.	Body-only policy is explicitly validated, not assumed.
B and C look similar	Wrong grade from shared appearance cues.	Use controlled mixed-grade validation; report B vs C confusion.	B/C confusion under same light/background.	This is a known hard boundary.
Strong glare / lighting change	Shininess/color features may change without quality change.	Do not let unvalidated appearance features override detector evidence; test acquisition conditions.	Same-fish lighting challenge; background-only/fish-only controls.	Appearance features are not causal proof.
Model 1 false candidate	Fake fish could enter grading pipeline.	Discard with validated verifier/ROI/temporal consistency only if tested; otherwise flag.	Model 1 FP study; verifier retention vs FP removal.	Verifier is only claimed after measured net benefit.
Model 2 finds no parts	Real fish may become evidence-poor.	Ungraded / re-observe across frames; do not force a quality class.	Ungraded reasons and coverage.	Abstention is deliberate.
Conflicting parts	Body/Head/Tail support different grades.	Use fixed weighted support + winner	Score margin/error curve.	Weighted support is not probability.

		margin; Ungraded if ambiguous.		
Edge-of-frame fish	Truncated anatomy and incomplete evidence.	Flag out-of-frame/low coverage; do not treat truncation as defect.	Edge subgroup performance.	Image truncation ≠ physical damage.
Unknown object resembles fish	Open-world false positive.	Target-domain constraints, hard negatives, optional verifier/unknown handling.	Hard-negative test set.	No model can guarantee rejection of every unseen object.
Video ID switch	Same fish counted twice or evidence mixed.	Track-level validation; use unique crossing logic; inspect ID switches.	ID-switch count and count MAE.	Tracking accuracy is separate from frame detection.
Same physical fish in multiple splits	Inflated validation/test performance.	Group by specimen/session; one physical fish = one split.	Specimen manifest and leakage audit.	Image-level split is insufficient if specimens repeat.
Rejected fish with one severe defect	Weighted average could hide critical defect.	Use expert-validated rejection override only when positive defect evidence exists.	Expert rubric + rejection recall/false acceptance.	Rejected requires positive evidence, not missingness.

5.2 Hard Questions Most Likely to Matter

Q1. Why do you need two models?

Short answer: Model 1 establishes the parent fish location/identity; Model 2 supplies grade-aware anatomical evidence. Separating them lets us evaluate parent detection, part detection, ownership, and final grading independently instead of hiding all errors inside one black-box output.

Avoid: Do not say two models are automatically more accurate.

Q2. Why 50/30/20?

Short answer: It is a current design hypothesis, not an industry standard. We independently ask the source-of-truth annotator to rank and allocate 100 points among Body/Head/Tail, then compare the expert-informed weights against alternatives on validation data before freezing the policy.

Avoid: Never say “because we decided the Body is more important.”

Q3. What if the Head belongs to another overlapping fish?

Short answer: The association layer treats ownership as a separate uncertainty problem. A part can remain ambiguous/unassigned rather than being forced to the nearest fish. Final grading can abstain if reliable evidence becomes insufficient.

Avoid: Do not claim association is perfect without parent-ID benchmark results.

Q4. Why not one end-to-end model?

Short answer: A one-model architecture is possible, but our current labels and thesis questions separate whole-fish identity from grade-part evidence. The two-stage design also gives more

interpretable failure analysis. We would simplify only if comparative validation shows equal or better end-to-end performance.

Avoid: Do not claim one model “cannot” solve the task.

Q5. Why not segmentation?

Short answer: Segmentation could help exact fish-only features and overlap handling, but it adds annotation/training complexity. We treat it as a measured-bottleneck solution, not a mandatory technology.

Avoid: Do not present segmentation as already implemented if it is not.

Q6. Why does Model 1 work on the conveyor but get confused elsewhere?

Short answer: That is a background/domain-shift issue. The correct response is to measure it, add representative background diversity and hard negatives, and define the intended operating environment.

Avoid: Do not say the system works on all backgrounds.

Q7. Is mAP50 0.79 the accuracy?

Short answer: No. mAP is an object-detection metric across localization/classification conditions. Final fish-level grading accuracy/F1 must be measured separately after association and decision fusion.

Avoid: Never label detector mAP as final classification accuracy.

Q8. YOLO has better mAP but Faster R-CNN has slightly better F1. Which is better?

Short answer: On the internal development set, YOLO has higher AP and much lower latency, while F1 is nearly tied and slightly favors Faster R-CNN at its own tuned threshold. We therefore call YOLO the current internal lead, not the final winner.

Avoid: Do not cherry-pick one metric without context.

Q9. Why did EfficientNet score zero?

Short answer: The tested EfficientNetV2-M + RetinaNet configuration failed to produce usable evaluated detections. That is a configuration/experiment failure requiring diagnosis, not proof that EfficientNet itself cannot work.

Avoid: Never say “EfficientNet is bad.”

Q10. What if the Tail is missing?

Short answer: First we must distinguish actual physical absence from occlusion, out-of-frame truncation, or detector failure. A missing detection alone does not justify Rejected.

Avoid: Do not map NOT_DETECTED directly to REJECTED.

Q11. What is the difference between Rejected and Ungraded?

Short answer: Rejected is a quality verdict supported by positive unacceptable-quality evidence. Ungraded is an uncertainty outcome when evidence is insufficient, ambiguous, or unreliable.

Avoid: Do not treat Ungraded as the worst quality class.

Q12. How do you know the model is learning fish rather than background?

Short answer: The final study should use specimen-safe splits, mixed B/C scenes under shared conditions, background-generalization tests, and ideally fish-only/background-only controls. These experiments test shortcut learning directly.

Avoid: Do not use internal mAP alone as proof of causal fish features.

Q13. Are handcrafted features actually used?

Short answer: Most appearance/geometry features are research/support features. We will only claim they improve grading if an ablation shows a measurable gain over Model 2-only evidence.

Avoid: Do not list every extracted feature as if it drives the runtime decision.

Q14. Could the same fish be in train and test?

Short answer: That is exactly why physical specimen IDs and session-aware grouping are required for the final split. Image-level duplicate checks alone are not enough.

Avoid: Do not say "no leakage" without specimen-level evidence.

Q15. What is your final accuracy?

Short answer: We will report final fish-level metrics only after the entire pipeline, thresholds and weights are frozen and evaluated once on an untouched specimen-independent test set. Current architecture numbers are internal-development evidence.

Avoid: Do not invent or substitute component metrics.

5.3 Live Demo Rescue Plan

If this happens	Do this	What to say
Camera/browser fails	Switch to pre-recorded demo or saved test images.	"The model pipeline is unchanged; we are using a fixed input because the live capture surface is unavailable."
Random object becomes a fish	Keep the example; explain the Model 1 false-positive stage and planned/validated mitigation.	"That is a Model 1 false positive, which we measure separately rather than hiding."
Real fish becomes Ungraded	Show the evidence coverage/reason code.	"The system intentionally abstains when evidence is insufficient instead of forcing a grade."
Wrong B/C grade	Show the confusion/failure analysis and controlled B/C validation plan.	"B/C is the hardest boundary; the thesis reports this limitation explicitly."
Overlap assigns wrong part	If available, show association/ambiguity output; otherwise admit the failure.	"Part ownership is a separate evaluated layer; heavy overlap is an identified stress case."

5.4 Phrases to Memorize

• “This is an internal-development result, not our final specimen-independent result.”

• “mAP evaluates object detection; final fish-grade performance is measured separately.”

• “Missing detection is not proof of physical absence.”

• “Rejected is positive bad-quality evidence; Ungraded is insufficient evidence.”

• “We do not renormalize missing evidence back to 100%.”

• “The 50/30/20 weighting is a hypothesis until expert validation and ablation support it.”

• “Ambiguous part ownership can remain Unassigned rather than being forced.”

• “We only claim feature benefit when an ablation demonstrates incremental value.”

• “The final test is run only after the model, thresholds, association and grading policy are frozen.”

6. Defense Readiness Checklist

Complete the P0 items before presenting final performance claims.

Priority	Checklist item	Why it matters	Status
P0	**Expert/annotator source-of-truth interview completed and archived.**	Backs grade definitions, rejection criteria, part importance and missing-part handling.	☐
P0	**50/30/20 weight ablation completed on validation set.**	Defends the weighting with evidence rather than opinion.	☐
P0	**Body-only/2-region/3-region gating policy decided from expert + validation evidence.**	Prevents arbitrary partial-evidence behavior.	☐
P0	**Specimen-aware final-test manifest frozen.**	Protects against data leakage accusations.	☐
P0	**End-to-end fish-level confusion matrix and macro F1 produced.**	Answers the real thesis outcome question.	☐
P0	**Rejected vs Ungraded semantics confirmed by expert.**	Prevents safety/quality-category confusion.	☐
P0	**Overlap/association benchmark completed or explicitly scoped as limitation.**	Defends part ownership in crowded scenes.	☐

P0	**Model/checkpoint/config hashes archived.**	Makes results reproducible and prevents model-version confusion.	☐
P0	**Claim → Evidence Matrix updated with final-test sources.**	Prevents overclaiming during Q&A.	☐
P0	**Pre-recorded demo + saved examples prepared.**	Protects against live-demo failure.	☐
P1	**Background/hard-negative benchmark completed.**	Quantifies the known false-positive/domain-shift issue.	☐
P1	**Verifier experiment completed OR verifier clearly removed from claimed system.**	Avoids presenting a proposed component as implemented.	☐
P1	**Model 2 crop vs full-frame integration comparison completed.**	Supports final integration choice.	☐
P1	**Model 2-only vs engineered-feature ablation completed.**	Proves whether handcrafted features add value.	☐
P1	**B/C controlled mixed-scene validation completed.**	Addresses likely shortcut/lighting confounds.	☐
P1	**End-to-end latency measured on defense/target hardware.**	Supports practical performance claims.	☐
P2	**Segmentation experiment performed only if background/overlap remains a bottleneck.**	Avoids unnecessary complexity.	☐

6.1 Defense Evidence Folder Structure

Recommended folder:

```
Defense_Evidence/
  01_system_source_of_truth/
  02_model_registry_and_hashes/
  03_expert_validation/
  04_weight_ablation/
  05_final_test_manifest/
  06_end_to_end_results/
  07_overlap_association/
  08_background_false_positives/
  09_B_vs_C_analysis/
  10_failure_examples/
```

11_demo_backup/
12_claim_evidence_matrix/

6.2 Final Team Alignment Exercise

Before the defense, every presenter should independently answer these without looking at notes. If answers differ, the source-of-truth document wins and the script/slides must be corrected.

☐ What does Model 1 do?

☐ What does Model 2 do?

☐ What are the 12 Model 2 classes?

☐ What are the current Body/Head/Tail weights?

☐ Why are missing weights not renormalized?

☐ What is the difference between Rejected and Ungraded?

☐ How are overlapping parts assigned?

☐ What is the strongest internal Model 2 result and why is it not final accuracy?

☐ What is the biggest current dataset validity risk?

☐ What result are we not yet allowed to claim?

Appendix A. Two-Minute Defense Story

A concise answer to “Explain your entire system.”

“Our system separates fish localization from quality evidence. Model 1 first detects each individual dried tamban and provides the parent fish region or track identity. Model 2 then detects grade-aware anatomical evidence for the Head, Body and Tail using 12 classes covering A, B, C and Rejected. Because multiple fish may overlap, part ownership is treated as a separate association problem; uncertain parts can remain unassigned instead of being forced to the nearest fish. The grading engine then combines only reliable evidence using fixed Body, Head and Tail weights and does not redistribute missing evidence. If support is insufficient or ownership is ambiguous, the system returns Ungraded rather than inventing a confident class. Rejected is reserved for positive unacceptable-quality evidence, not mere nondetection. Our current detector metrics are internal-development evidence. The final thesis result is the end-to-end fish-level evaluation on a locked, specimen-independent test set after expert validation and ablation of the grading policy.”

Appendix B. One-Page Source-of-Truth Snapshot

Model 1	Whole-fish parent detector/localizer; supports counting/tracking.
Model 2	12 grade-part detections: A/B/C/Rejected × Body/Head/Tail.

Current Model 2 lead	YOLO26m on internal development: mAP50-95 0.4606, mAP50 0.7876, P 0.8018, R 0.8625, F1 0.8310, ~54.6 ms/image.
Association	Part → parent Fish ID; allow ambiguous/unassigned.
Weights	Current hypothesis: Body 0.50, Head 0.30, Tail 0.20.
Missing evidence	Not renormalized.
Rejected	Positive evidence of unacceptable quality.
Ungraded	Insufficient/ambiguous/unreliable evidence.
Handcrafted features	Research/support features; benefit not claimed without ablation.
Verifier	Proposed false-positive mitigation; not baseline until evaluated.
Main scientific risks	Specimen independence, B/C confounding, overlap ownership, background false positives, unvalidated weights/gates.
Final thesis metric	End-to-end fish-level A/B/C/Rejected performance + Ungraded coverage on unseen specimen-independent test data.

Appendix C. Decision Log

Decision	Options considered	Evidence used	Final choice	Date/owner
Final Model 2 architecture	_____	_____	_____	_____
Part-weight policy	_____	_____	_____	_____
Minimum evidence gate	_____	_____	_____	_____
Association thresholds	_____	_____	_____	_____
Rejected override rules	_____	_____	_____	_____
Final runtime integration path	_____	_____	_____	_____
Verifier inclusion	_____	_____	_____	_____

Final test release	_____	_____	_____	_____

Appendix D. Evidence Provenance

This defense pack consolidates the current project evidence developed across the repository audit, Model 1/Model 2 technical documentation, the three-model Model 2 internal-development experiment, the expert/annotator validation protocol, and the thesis claim audit. Where a result has not yet been produced, the document intentionally uses a blank field or “pending” status rather than inventing evidence.

• Current/most recent repository audit and code review.

• Model 2 r03 three-model internal-development comparison (YOLO26m, Faster R-CNN R50-FPN V2, EfficientNetV2-M RetinaNet).

• Two-model technical documentation and association/grading roadmap.

• Expert/Annotator Validation Protocol.

• Thesis Claim Audit and manuscript consistency review.